

AIKA ONO

Medical Student · Aspiring Design Engineer

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PROFILE

Medical student moving into design engineering. I want to make things that are beautiful, that hold both functionality and human warmth, and that give people a sense of awe. Interests in parametric / generative design, IoT, and biomimetics.

EDUCATION

Imperial College London

2025 – 2026

iBSc Biomedical Engineering — First Class Honours

- **Core modules:** Fundamentals of Biomedical Engineering; Probability, Statistics and Data Analysis; Journal Club.
- **Optional modules:** Biomedical Instrumentation; Orthopaedic Biomechanics; Biomimetics; Advanced Physiological Monitoring and Data Analysis.

University of Edinburgh

2023 – 2029

MBChB Medicine

Pre-university

- **A-levels:** A*A*A*A in Mathematics, Further Mathematics, Chemistry, EPQ, and Biology.
- **GCSEs:** 11 × Grade 9.
- **Admissions tests:** BMAT 7.5 / 6.8 / 3A; UCAT 3120 (Band 1).

SELECTED PROJECTS

Linearity Index — Computational Design Tool

Solo

- Built and deployed a web tool that quantifies how linear or curved a form is: it samples an object's outline, measures curvature, and returns a single score from 0 to 1.
- Places any form on a spectrum from organic (an Alvar Aalto vase) to rectilinear (Le Corbusier furniture) — making an aesthetic property measurable.

Soft-Robotic Hydraulic Gripper

Group

- Hydraulic-actuated silicone gripper able to grasp delicate objects with closed-loop control. Responsible for design and manufacturing.

Additional making projects

- **Keshiki** — IoT device that senses and expresses emotional state.
- **Mechanical clock redesign** — mechanism reworked using fluid-dynamics analysis.

EXPERIENCE

NI Drive

June 2026 – Present

Engineering Intern

Systems Biology Lab — University of Edinburgh

Summer 2024

- Built neural networks (including an LLM) and mathematical models to predict gene expression directly from DNA sequence, extracting patterns from large genomic datasets in Python.

AWARDS & PRESENTATIONS

- **Best Use of Atech** — Oxford AI and hardware hackathon.
- **Poster presentation** — “AI-Powered Virtual Artists: Revolutionising Intraoperative Image-Guided Neurosurgery,” RiiSE24 and the Royal College of Surgeons' Robotics for Trainees.
- **Silver medal** — **iGEM** (*aLFA reMask*). Synthetic-biology team project exploring the interface between biology and hardware.

TECHNICAL SKILLS

Design: Fusion 360, Rhino / Grasshopper, OpenSCAD

Fabrication: 3D printing, laser cutting, silicone casting

Electronics: Arduino, ESP32

Programming: Python, machine learning, data analysis

Visual design: Clip Studio Paint, Adobe Illustrator

LEADERSHIP & INTERESTS

Roles: Publicity Coordinator, Radiology Society; Voted General Member, MedTech Society

Interests: Illustration (traditional and digital), hip-hop dance, philosophy, poker

Languages: English (fluent), Japanese (fluent)